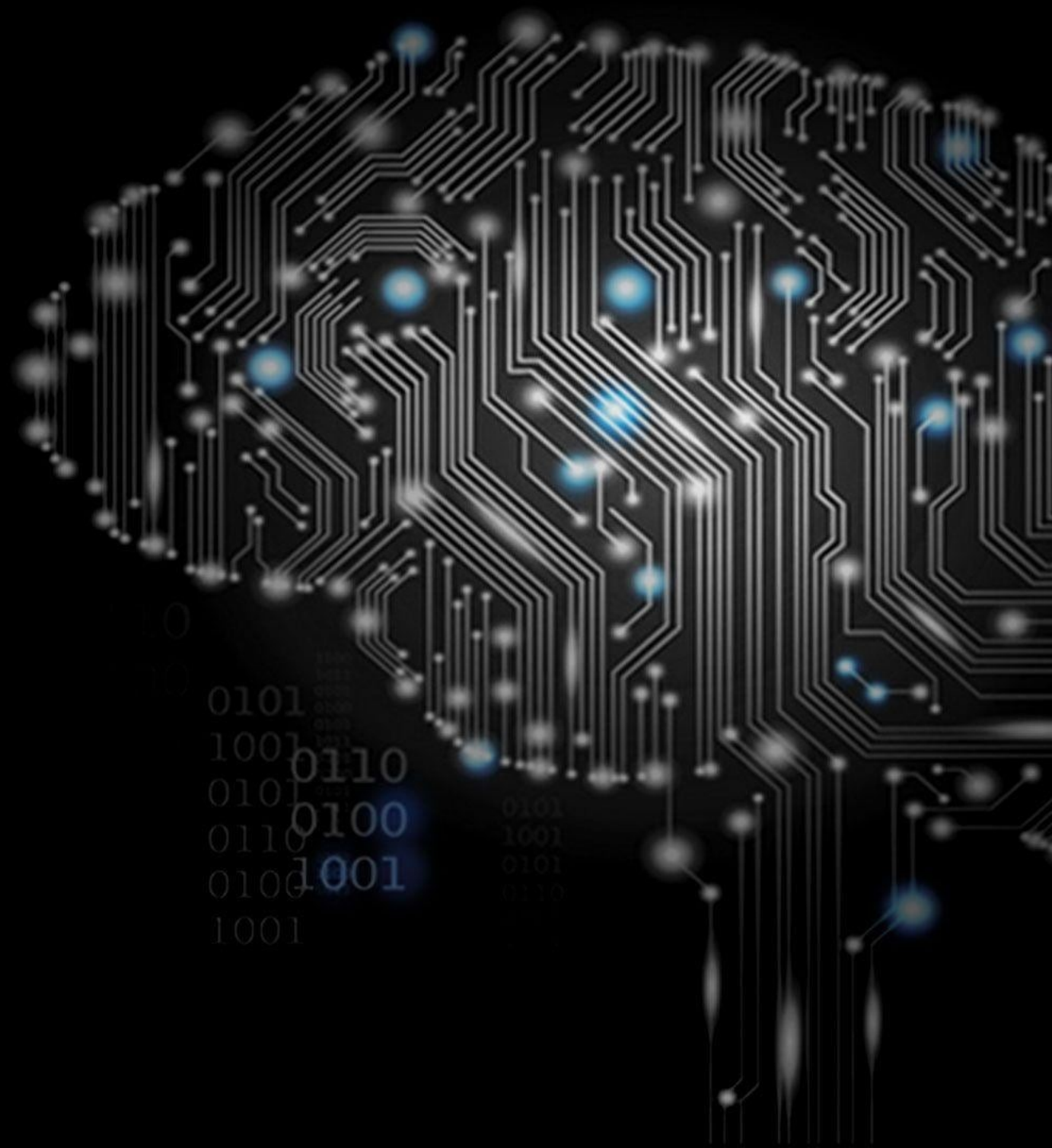


Introduction to Generative AI

Course Introduction



Course Introduction

- The field of generative AI (GAI) and other AI-related technologies are in a constant state of churn
 - New developments, techniques, and technologies are rapidly being developed and deployed
 - This has resulted in a lot of hype and misinformation from a variety of sources
- This course is designed to provide a solid technical overview of GAI as of 2026
 - The course is not a technical deep dive, but a survey of the major topics that are current in the field of GAI
 - The objective of this course is to introduce the main GAI technologies that software developers and engineers need to be familiar with



Course Topics

- The AI Landscape
- Transformers and Large Language Models
- Diffusion Models
- Prompt Engineering
- Retrieval-Augmented Generation (RAG)
- Fine-Tuning and Model Adaptation
- AI Agents and Tool-Using Models
- Evaluation, Safety and Governance



Logistics

- Repository for class materials
 - <https://github.com/ExgnosisClasses/2610-GenAI-March-17>
- Hours 9am – 5pm CST, March 17-18, 2026
 - Lunch is 12-1 CST
 - Two 15 breaks, one mid-morning and one mid-afternoon
- Labs repository
 - <https://github.com/ExgnosisClasses/2610-GenAi-Labs>
- Lab machines
 - VMs will be provided for the labs
 - Lab #1 will walk you through how to log on to your VM



Instructor – Rod Davison

- 50+ years experience in IT
- 40+ years experience in AI
 - Designed and developed AI prototypes
 - NLU, expert systems, machine translation, agents and more
 - Research director for AI companies
- Academic background
 - Theoretical math, psychology, anthropology, cognitive science
- Consulting
 - Finance, Defence, Insurance, Government, Fortune 100
 - Multiple areas



Introductions

- In order for me to ensure the class meets your needs
- Please introduce yourself and tell me:
 - Your professional background and skill set (developer, BA, etc.)
 - Your experience in using AI and GAI
 - Any specific goals or objectives for the class



Questions?

